

COL 703/7203

8th October

Claim: For a function $f: \{0,1\}^n \rightarrow \{0,1\}$
there is exactly one ROBDD u
with ordering $x_1 < x_2 < \dots < x_n$ such that
 u represents $f(x_1, x_2, \dots, x_n)$.

Proof: Induction on the number of
parameters (n).

Exercise

COL703/7203: Logic for Computer Science (Jul-Nov 2026)

Predicate Logic

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The need for a richer language

- the logical aspects of natural and artificial languages are much richer
 - than what propositional logic can capture
 - limited to sentence components like not, and, or, if ... then

- consider the following declarative sentence

Every student is younger than some instructor.

- a propositional atom denoting this fails to capture the finer logical structure of this sentence.

The need for a richer language

Every student is younger than some instructor.

this is about being a student, being an instructor, and being younger than somebody else

we would like a mechanism to express these with their logical relationships

this is what we will use **predicates** for

Predicates

- we can use predicates S , I , and Y
- $S(\text{John})$ – John is a student
- $I(\text{Paul})$ – Paul is an instructor
- $Y(\text{John}, \text{Paul})$ – John is younger than Paul
- but the meaning of these (predicate) symbols must be specified exactly

Predicates

- we don't want to write down every instance of $S(.)$
- use **variables** as place holders for concrete values
- $S(x)$ – x is a student
- $I(x)$ – x is an instructor
- $Y(x, y)$ – x is younger than y

Quantifiers

- can we now express "Every student is younger than some instructor."

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- $\forall$ (for all) and $\exists$ (there exists)

Quantifiers

- can we now express "Every student is younger than some instructor."
- Every student is younger than some instructor.
- $\forall$ (for all) and $\exists$ (there exists)
- the quantifiers always come attached to a variable
- $\forall x$ (for all x) and $\exists z$ (there exists z)

Example

- can we now express "Every student is younger than some instructor."

Example

- can we now express "Every student is younger than some instructor."
- $\forall x (S(x) \rightarrow (\exists y (I(y) \wedge Y(x, y))))$
- for every x , if x is a student, then there is some y such that y is an instructor and x is younger than y
- predicates can have any (finite) number of arguments (arity)

Another example

- Not all birds can fly.
- $B(x)$ – x is a bird
- $F(x)$ – x can fly

$$\exists x (B(x) \wedge \neg F(x))$$

It is not the case that
all birds can fly.

$$\neg \forall x (B(x) \rightarrow F(x))$$

Another example

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- $\neg(\forall x (B(x) \rightarrow F(x)))$

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- alternatively, $\exists x (B(x) \wedge \neg F(x))$

Another example

- Not all birds can fly.
- $B(x)$ – x is a bird
- $F(x)$ – x can fly
- $\neg(\forall x (B(x) \rightarrow F(x)))$
- alternatively, $\exists x (B(x) \wedge \neg F(x))$
- does this formula evaluate to true in the world we currently live in?

Another example

Every child is younger than its mother.

Another example

Every child is younger than its mother.

$$\forall x \forall y ((C(x) \wedge M(y, x)) \rightarrow Y(x, y))$$

Another example

Andy and Paul have the same maternal grandmother.

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Andy and Paul have the same maternal grandmother.

$$\forall x \forall y \forall u \forall v ((M(x, y) \wedge M(y, \text{Andy}) \wedge M(u, v) \wedge M(v, \text{Paul})) \rightarrow x = u)$$

$$\exists x \exists y \exists v (M(x, y) \wedge M(y, \text{Andy}) \wedge M(x, v) \wedge M(v, \text{Paul}))$$

Another example

Andy and Paul have the same maternal grandmother.

$$\forall x \forall y \forall u \forall v ((M(x, y) \wedge M(y, \text{Andy}) \wedge M(u, v) \wedge M(v, \text{Paul})) \rightarrow x = u)$$

- equality has been used as a special predicate
- function symbols can help us avoid the inelegant encoding
- $\forall x (C(x) \rightarrow Y(x, m(x)))$
- $m(m(\text{Andy})) = m(m(\text{Paul}))$

As a formal language

there are two sorts of things involved in a predicate logic formula:

objects that we are talking about – constants, variables, $m(a)$, $g(x, y)$

expressions denoting objects are called terms

the other sort of things are formulas

Terms

(Syntax of Predicate logic)

↳ signature (constants, function symbols, predicate symbols)

- every variable is a term $\boxed{t ::= x \mid c \mid f(t_1 \dots t_n)}$
- If c is a nullary function (i.e., a constant symbol) then c is a term.
- If $t_1, \dots, t_n$ are terms and f is an n -ary function symbol then $f(t_1, \dots, t_n)$ is a term.
- Nothing else is a term.

Formulas

- If P is a predicate symbol of arity $n \geq 0$
and if $t_1 \dots t_n$ are terms then
 $P(t_1, \dots, t_n)$ is a formula.
- If ϕ is a formula $\neg \phi$ is a formula.
- If ϕ and ψ are formulas $\phi \vee \psi$, $\phi \wedge \psi$, $\phi \rightarrow \psi$
are formulas.
- If ϕ is a formula and x is a variable
then $\forall x \phi$ and $\exists x \phi$ are formulas.
- Nothing else is a formula.

Binding properties

- $\neg$, $\forall y$, $\exists y$ bind most tightly
- then $\vee$ and $\wedge$
- then $\rightarrow$, which is right-associative

Another example

Every son of my father is my brother.

$\forall x (S(x, f(m)) \rightarrow B(x, m))$

m constant symbol
(for myself)

$F(x, y)$

x is the father
of y

$B(x, y)$

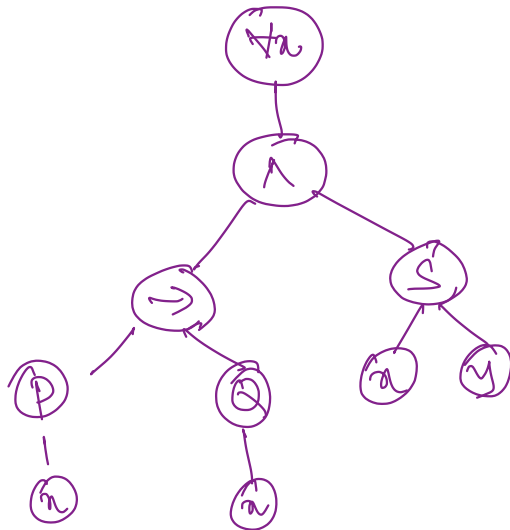
x is a brother
of y

$S(x, y)$

x is a son of y .

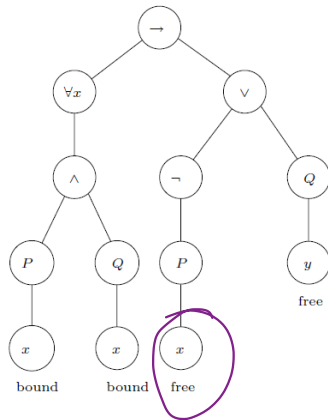
Parse trees

$$\forall x ((P(x) \rightarrow Q(x)) \wedge S(x, y))$$



Free and bound variables

$$(\forall x (P(x) \wedge Q(x))) \rightarrow (\neg P(x) \vee Q(y))$$



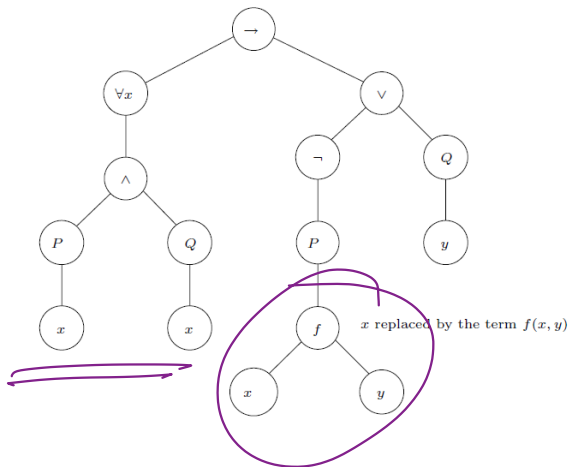
Substitution

given a variable x , a term t , and a formula ϕ

we define $\phi[t/x]$ to be the formula obtained by replacing each **free occurrence** of variable x in ϕ with t

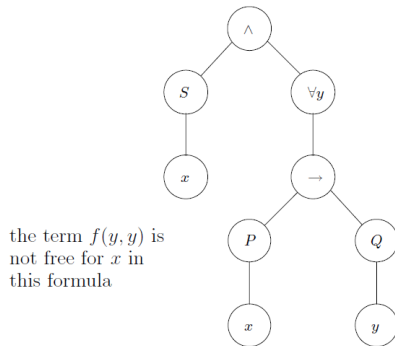
Substitution: example

example: $((\forall x (P(x) \wedge Q(x))) \rightarrow (\neg P(x) \vee Q(y)))[f(x,y)/x]$



Undesired side-effects of substitution

substitution must be **avoided** if t is not free for x in ϕ



given a term t , a variable x , and a formula ϕ , we say that **t is free for x in ϕ** if no free x leaf in ϕ occurs in the scope of $\forall y$ or $\exists y$ for any variable y occurring in t

Proof theory (natural deduction rules)

Proof rules for equality

$$\frac{}{t = t} = i \quad (\text{axiom})$$

$$\frac{t_1 = t_2 \quad \phi[t_1/a]}{\phi[t_2/a]} = e$$

Exercise

$$t_1 = t_2 \quad \vdash \quad t_2 = t_1$$

$$t_1 = t_2, t_2 = t_3 \quad \vdash \quad t_1 = t_3$$